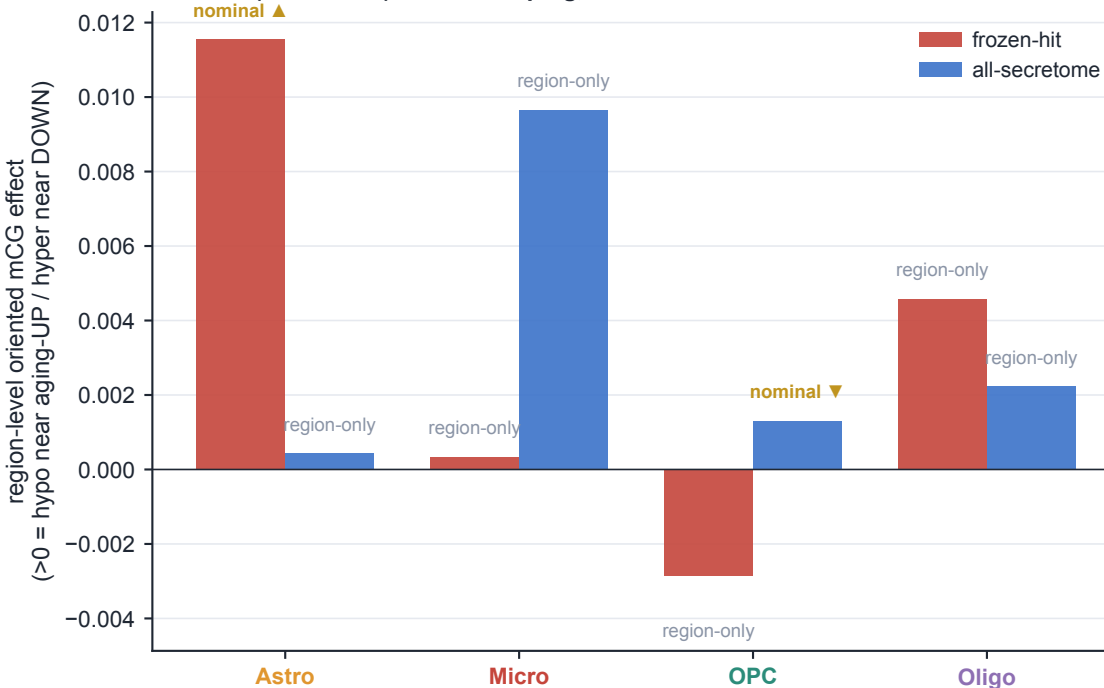


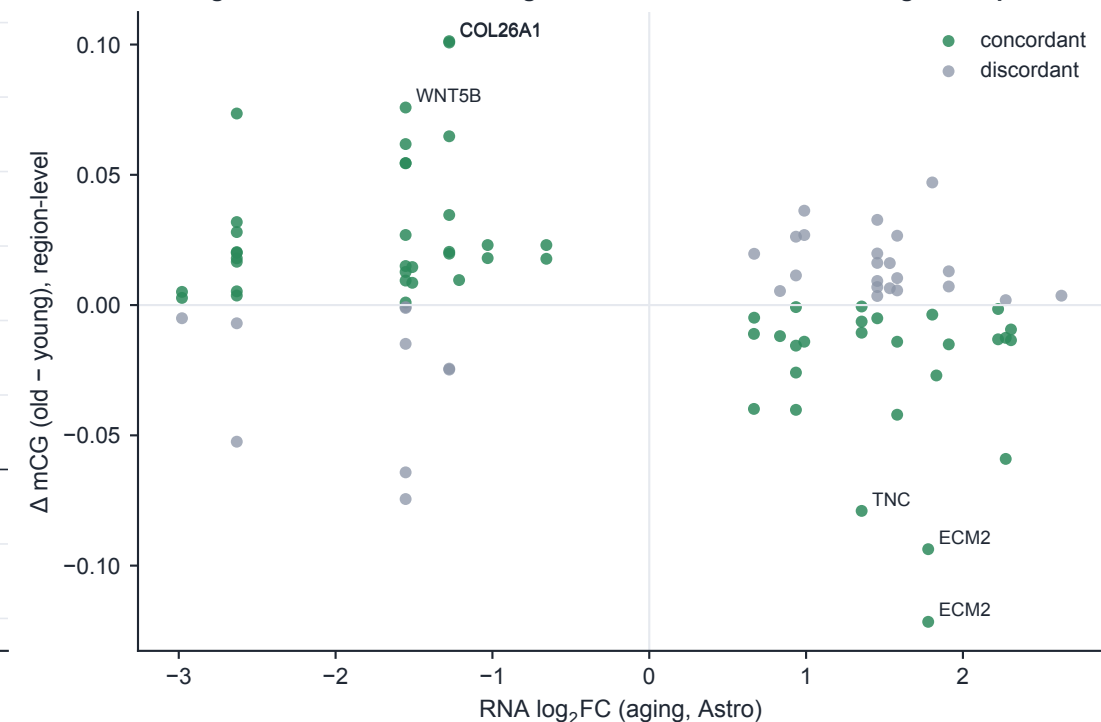
Figure S-M1 · mCG age-dynamics over secretome regulatory regions (GSE299139 snm3C, 40 donors, hg38) — DESCRIPTIVE

The RNA-directed coordination reading is **WITHDRAWN**: the one nominal positive (Astro frozen-hit) fails four controls;
the methylome layer is reported as a directionally non-specific mCG bias and is **NOT** counted as regulatory support.

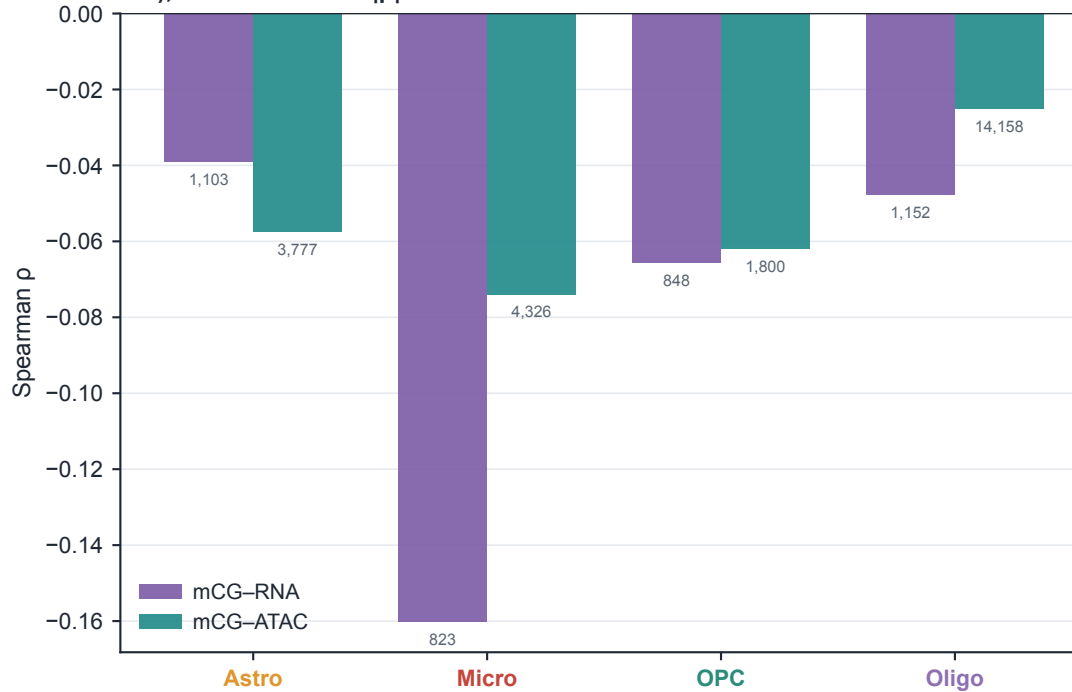
A Secretome regulatory regions (region-level bars; descriptive)
nominal = gene-level matched-null AND sign-perm $p < 0.05$ BEFORE the four controls (▲ ▼ below) — bookkeeping, not a robustness verdict



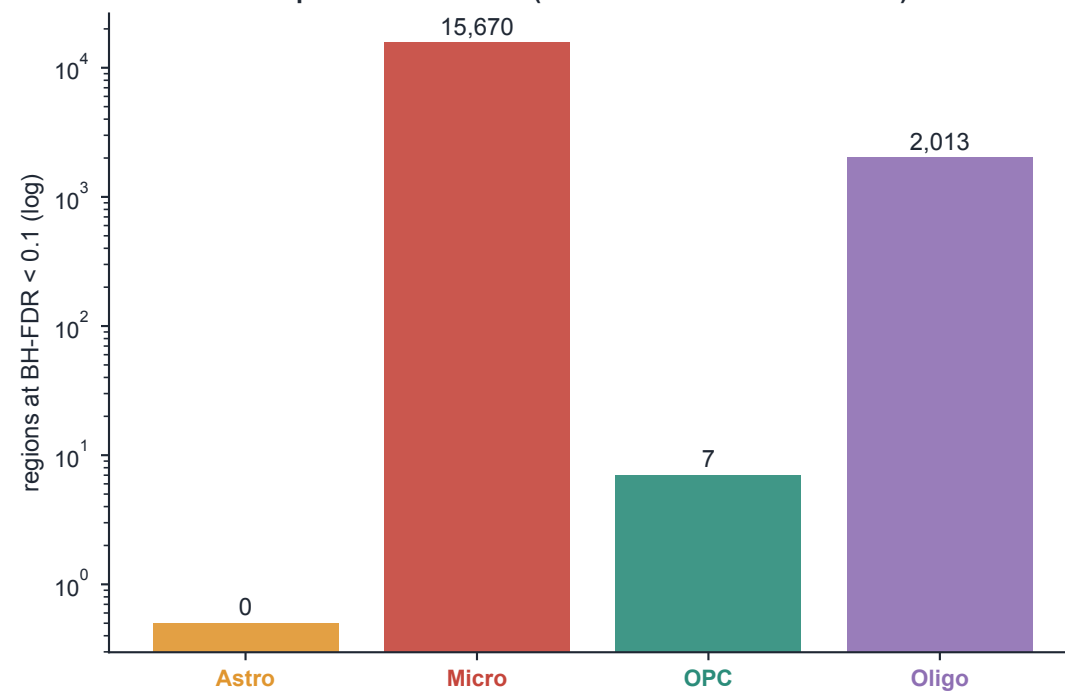
B Astro frozen-hit regions — region-level UNADJUSTED reference (n = 92 regions, 65% concordant)
aging-UP→hypomethylation, aging-DOWN→hypermethylation. Covariate-unadjusted, and regions are correlated within genes — NOT on the same footing as A's p-values.



C Cross-modal effect-size correlation (cross-cohort, NOT donor-paired)
mCG[GSE299139] vs RNA/ATAC[GSE268609]; mCG↔ATAC inverse expected by construction.
Descriptive: ρ is the result, no p shown — rows are genes/peaks (n under each bar), not donors. max $|\rho| = 0.16$.



D Genome-wide age-methylation (FDR<0.1; context only)
Micro NOT composition-controlled (Micro1 absent in 6 old donors)



▲ Astro frozen-hit (gene-level $p_{\text{matched}} 0.018$ / $p_{\text{signperm}} 0.043$) FAILS all four pre-submission controls — substate pooling (Astro1 alone 0.836, Astro2 alone 0.291); orientation by another cell type's RNA does not weaken it (OPC orientation 0.019 / 0.025); non-secretome RNA-significant genes match it (0.023, $n = 155$); indistinguishable from a 10,000× scrambled-orientation null (empirical $p = 0.333$). → the RNA-directed coordination reading is **WITHDRAWN**; this layer is not regulatory support.

▼ OPC broad set = 0.31 percentage points of methylation over the 51-year span — below what we consider biologically interpretable.